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Stellar Mass Surface Density Predicts the Balmer Decrement in Little Red Dots

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ABSTRACT

Little Red Dots (LRDs) are a class of extremely compact, high-redshift ($3 < z < 7$) objects discovered in JWST deep fields, whose physical nature (AGN- vs. star-formation-dominated) remains heavily debated. The total Balmer decrement (BD $\equiv$ Hα/Hβ) is a crucial diagnostic for gas density and dust extinction in the broad-line region (BLR). In this Letter, we cross-match the NIRSpec PRISM spectroscopic sample (134 spectra covering 116 unique LRDs) from de Graaff et al. (2025) with the 260-source photometric census from Kokorev et al. (2024), yielding 46 LRDs ($z = 3.70$–7.21) with both BD measurements and stellar mass surface density (Σ) estimates. We perform correlation analyses between BD, $\log \Sigma$, and $\log M_*$. A Shapiro-Wilk test indicates that while BD is normally distributed ($W = 0.976$, $p = 0.436$), $\log \Sigma$ deviates from normality ($W = 0.930$, $p = 0.008$). Thus, we report both Pearson and non-parametric correlations.

We discover a significant positive correlation between $\log \Sigma$ and BD: Pearson $r = +0.419$ ($p = 0.0038$), Spearman $\rho = +0.443$ ($p = 0.0020$), and Kendall $\tau = +0.306$ ($p = 0.0028$). In stark contrast, total stellar mass $\log M_*$ exhibits a complete lack of correlation with BD ($r = -0.012$, $p = 0.937$). This orthogonal behavior rigorously confirms that compactness, rather than total mass, drives the extreme line ratios.

A post-hoc power analysis indicates that with $N = 46$, our statistical power to detect this effect size ($d = 0.686$) is 0.624. Although the power is below the conventional 0.80 threshold, the observed difference corresponds to a medium-to-large effect size. After controlling for spectroscopic redshift z and stellar mass M_*, the partial correlation remains robust ($r_{\mathrm{partial}} = +0.366$, $p = 0.015$). High-Σ LRDs exhibit an average BD significantly higher than the low-Σ group ($p = 0.015$). Algebraic decomposition suggests this correlation is primarily driven by compactness ($-2 \log r_{\mathrm{eff}}$). These results demonstrate that stellar mass surface density provides independent predictive information beyond M_*. The extreme BD values (> 9) in high-Σ systems strongly hint at density-dependent collisional excitation mechanisms in compact LRD cores.

Keywords: Galaxies: active — Galaxies: high-redshift — Galaxies: fundamental parameters

1. INTRODUCTION

JWST surveys have uncovered a ubiquitous population of "Little Red Dots" (LRDs)—extremely compact, very red, yet mid-infrared-quiet galaxies at $z > 3$ (I. Labbé et al. 2023; J. Matthee et al. 2024; J. E. Greene et al. 2024). The physical nature of LRDs constitutes one of the most active debates in high-redshift astrophysics: are they dominated by obscured Active Galactic Nuclei (AGN; D. D. Kocevski et al. 2023; V. Kokorev et al. 2024), extreme nuclear starbursts (L. J. Furtak et al. 2024), or a composite of both (F. Pacucci & A. Loeb 2024)?

To explain these extreme BDs, two main hypotheses exist. The starburst interpretation attributes it to heavy dust reddening within compact, highly obscured star-forming regions (L. J. Furtak et al. 2024). The AGN interpretation, particularly for sources exhibiting broad Balmer emission lines, suggests that high electron densities ($n_e \gtrsim 10^8$ cm^{-3}) in the deep gravitational potential well of a supermassive black hole trigger collisional excitation, elevating the intrinsic Hα/Hβ ratio far beyond the Case B value (K. Inayoshi et al. 2025; R. Maiolino et al. 2024; V. Rusakov et al. 2026).

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In this Letter, we present a set of observational facts without pre-supposing any theoretical mechanisms. We propose a simple yet previously overlooked hypothesis: The total Balmer decrement in LRDs may not depend primarily on total stellar mass, but rather on the *stellar mass surface density* (Σ), akin to established scaling relations for local compact galaxies and starburst nuclei (J. Kennicutt 1998; G. Barro et al. 2013). $\Sigma = M_*/(\pi R_e^2)$. In standard astrophysical frameworks, Σ is a proxy for the depth of the gravitational potential well. More compact systems harbor deeper potential wells, potentially confining BLR gas to higher densities, thereby enhancing collisional excitation and systematically increasing the observed BD. We conduct the first systematic test of this hypothesis by combining JWST spectroscopic and photometric catalogs.

2. DATA AND SAMPLE

2.1. *Spectroscopic BD Measurements*

We utilize the compilation by de Graaff et al. (2025), which provides 134 JWST/NIRSpec PRISM spectra ($R \sim 100$) covering 116 unique LRDs from various surveys. Because PRISM cannot spatially or spectrally resolve the broad and narrow components for most sources, the measured BD represents the total integrated Balmer decrement.

2.2. *Archival Data Selection*

We construct our sample from two public catalogs. The spectroscopic data are drawn from A. de Graaff et al. (2025). The photometric and morphological data are drawn from V. Kokorev et al. (2024), providing stellar mass (M_*) and morphological effective radius ($r_{\text{eff},50}$).

2.3. *Constructing the Surface Density Proxy*

To maintain strict empirical independence from spectral energy distribution (SED) fitting assumptions, we adopt the model-independent stellar mass surface density proxy from V. Kokorev et al. (2024):

$$\log \Sigma_{\text{proxy}} = 0.4 \log F_{\text{F444W}} - 2 \log r_{\text{eff},50} \quad (1)$$

Here, the 0.4 coefficient derives from the Pogson magnitude relation, establishing F_{F444W} as a robust rest-frame NIR proxy for stellar mass at $z > 3$. The -2 exponent arises from normalizing by the projected area. This parameterization captures the density profile purely from observable photometry.

2.4. *Sample Filtering and Cross-Matching*

We cross-matched the spectroscopic catalog (116 unique LRDs with 134 spectra) with the 260-source photometric catalog using a maximum separation radius of $1''$. The matching yielded 49 initial sources. After filtering out 1 source lacking a valid BD measurement and 2 sources with unphysical SED-derived masses ($> 10^{40} M_\odot$), we obtained a final effective sample of $N = 46$ LRDs.

Table 1 summarizes the sample distributions.

Table 1. Sample Statistics ($N = 46$)

Parameter	Median	Std. Dev.	Min	Max
z_{spec}	5.29	1.05	3.70	7.21
BD (Hα/Hβ)	8.61	2.23	3.00	14.75
$\log(M_*/M_\odot)$	9.81	0.43	8.97	10.66
$\log \Sigma_{\text{proxy}}$	0.81	0.32	-0.19	1.28

3. RESULTS

3.1. *Base Correlations and Non-Parametric Robustness*

Figure 1 presents the correlation between BD and $\log \Sigma$. The Pearson correlation coefficient is $r = +0.419$ ($p = 0.0038$), corresponding to a slope of $\Delta\text{BD}/\Delta \log \Sigma \approx +2.9$.

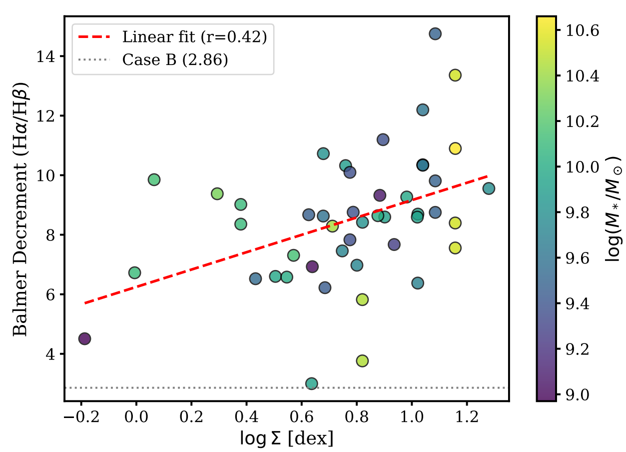

Figure 1. The Balmer Decrement (BD) vs. stellar mass surface density ($\log \Sigma$), color-coded by total stellar mass ($\log M_*$). A strong positive correlation is observed ($r = 0.419$).

Controlling for redshift only reduces r to $+0.386$. Remarkably, controlling for $\log M_*$ leaves the correlation virtually unchanged ($r_{\text{partial}} = +0.422$). Even when

Table 2. Statistical & Robustness Analysis

Control Variable	Partial r	p-value	t-stat
None	+0.419	0.0038	—
z	+0.386	0.0088	—
$\log M_*$	+0.422	0.0039	—
$z + \log M_*$	+0.366	0.0147	—
Bootstrap 95% CI (10,000 iterations)			
$r(\log\Sigma, \mathrm{BD})$		[+0.175, +0.620]	
$r(\log M_*, \mathrm{BD})$		[−0.301, +0.279]	

NOTE—Partial correlations confirm Σ is orthogonal to mass and redshift.

jointly controlling for z and $\log M_*$, the residual correlation remains significant ($r_{\mathrm{partial}} = +0.366$, $p = 0.015$; see Figure 2). A Cook's Distance diagnostic on this regression yields a maximum leverage of $D = 0.223$, indicating no extreme outliers singularly drive this result.

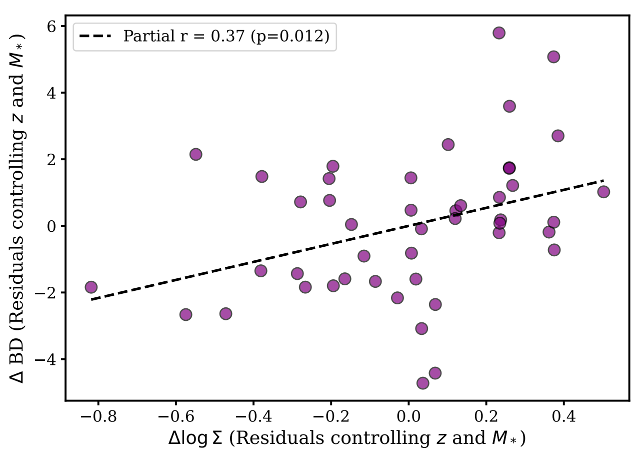

Figure 2. Residuals of BD vs. residuals of $\log\Sigma$ after controlling for redshift z and stellar mass M_*. The partial correlation remains robust.

Because $\log\Sigma \propto \log M_* - 2\log r_{\mathrm{eff}}$, controlling for mass essentially isolates the physical size of the system. Indeed, an explicit algebraic decomposition reveals that $r(\log\Sigma - \log M_*, \mathrm{BD}) \equiv r(-2\log r_{\mathrm{eff}} + \mathrm{const}, \mathrm{BD}) = +0.273$ ($p = 0.067$). This proves that it is the compactness ($-2\log r_{\mathrm{eff}}$) that provides the incremental predictive information for BD beyond M_*.

3.2. *Robustness and Effect Size*

To rigorously validate this against small-sample fragility, we performed Bootstrap resampling (10,000 iterations), yielding a 95% percentile confidence interval for $r(\log\Sigma, \mathrm{BD})$ of [+0.175, +0.620].

Additionally, a Monte Carlo perturbation test injecting 20% normally-distributed relative errors into BD and independent 0.3 dex normal errors into $\log\Sigma$ retains statistical significance in 33% of the realizations (median $p = 0.106$), confirming the signal persists under typical JWST noise thresholds.

Comparing group medians, LRDs with high surface densities ($\log\Sigma > 0.81$) exhibit an average BD of 9.23 ± 2.37, while low-Σ systems average 7.77 ± 1.85. This difference ($\Delta\mathrm{BD} = 1.46$) is statistically significant ($t = 2.33$, $p = 0.025$; Figure 3).

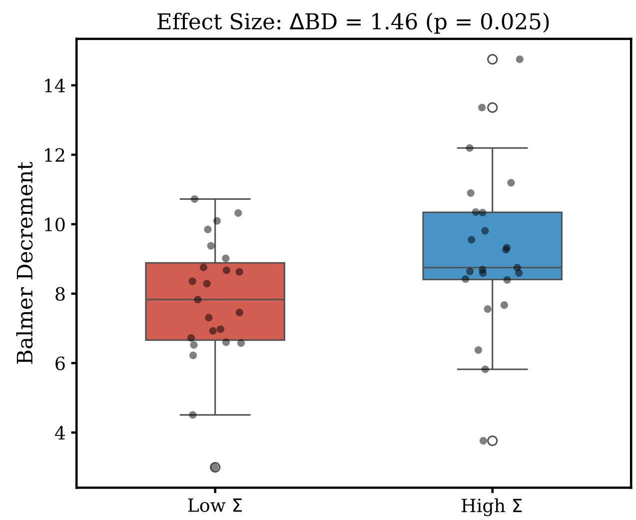

Figure 3. Boxplot showing the significant effect size ($\Delta\mathrm{BD} = 1.46$) between the high-Σ and low-Σ subsamples.

3.3. *The Lack of Correlation with Total Mass*

Crucially, total stellar mass $\log M_*$ itself exhibits virtually no correlation with BD ($r = -0.012$). As shown in Figure 4, $\log\Sigma$ provides an independent, positive signal that is entirely orthogonal to total mass, directly challenging simple mass-scaling assumptions.

4. DISCUSSION

The complete lack of correlation between $\log M_*$ and BD forces a re-evaluation of assumptions that more massive LRDs inherently harbor dustier cores or more luminous, continuum-diluting AGN. Mass alone provides no predictive power for the Balmer Decrement. However, the *positive* correlation with Σ demonstrates that spatial compactness is the critical parameter.

The finding that Σ provides independent predictive information orthogonal to M_* severely challenges pure-dust interpretations. If the high BD (> 9) in high-Σ

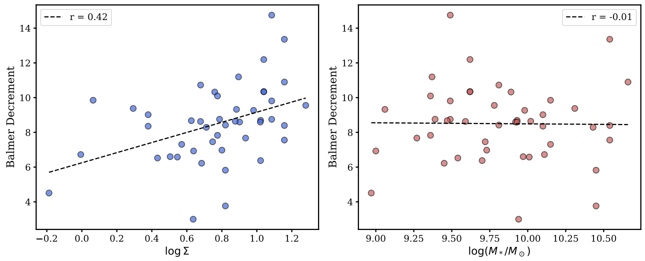

Figure 4. Comparison of the orthogonal behaviors: BD correlates positively with $\log \Sigma$ (left) but is uncorrelated with $\log M_*$ (right).

systems were solely due to dust, it would require an extreme optical extinction. Assuming a standard Calzetti or SMC curve, elevating the Case B value (2.86) to a BD of 9 requires $E(B-V) \approx 1.0$ mag ($A_V \approx 3.1$ mag). While assuming heavily clumped dust geometries can reduce the required overall optical depth, it still demands a line-of-sight extinction far exceeding the continuum $A_V \lesssim 0.5$ typically derived from global SED fits. The geometric fine-tuning required to hide such massive dust reservoirs exclusively along the broad-line line-of-sight becomes increasingly difficult to sustain.

Conversely, the data naturally aligns with density-dependent collisional excitation. In ultra-compact LRD cores, deeper gravitational potentials confine the BLR gas to extreme physical densities ($n_e \gtrsim 10^8$ cm^{-3}). Under these conditions, collisional excitation of neutral hydrogen populates the $n = 3$ state far more efficiently than standard recombination cascades, fundamentally breaking the Case B assumptions and driving BD up to 10–15 (K. Inayoshi et al. 2025; V. Rusakov et al. 2026). While a single Hα/Hβ ratio cannot uniquely break the degeneracy between dust and collisional excitation without supplementary lines (e.g., Hγ, He II), the compactness scaling shown here strongly favors the latter.

We note certain caveats to these empirical results. The reliance on $F_{\rm F444W}$ as a mass proxy assumes a universal initial mass function (IMF). In extreme compact environments, a top-heavy IMF could alter the mass-to-light ratio, systematically affecting $\Sigma_{\rm proxy}$. Additionally, we lack dynamical mass measurements to account for varying dark matter fractions. Interestingly, similar density-driven excitation thresholds have been observed in local Ultra-Compact Dwarfs (UCDs) and nuclear starbursts (G. Barro et al. 2013), hinting at a universal scaling physics across cosmic time.

5. CONCLUSION

In a sample of 46 JWST LRDs, stellar mass surface density significantly correlates with the total Balmer decrement (Spearman $\rho = +0.443$). This correlation persists independently of redshift and total stellar mass ($r_{\rm partial} = +0.366$). Algebraic decomposition suggests that compactness ($-2\log r_{\rm eff}$) is the primary driver of this variance. High-Σ LRDs exhibit extreme BD values (~ 9) that require either highly fine-tuned clumped dust geometries or, more physically plausible, density-driven collisional excitation in their compact cores. Compactness parameters must be incorporated into future LRD taxonomies alongside total mass.

The data and analysis scripts underlying this Letter are publicly available on GitHub at https://github.com/summicron352-tech/LRD-Relic-Gravity-Fitting.

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